SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0701

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks:50**

Annexure A

Scenario-Based

Code of Conduct:

I ___Sk. Parvez Ahamed (192411084)__ certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Sk. Parvez Ahamed*

1. Smart Campus Network — 192.168.100.0/24

The given network is 192.168.100.0/24, which contains 256 total IP addresses. The subnets are allocated using VLSM (Variable Length Subnet Masking).

Subnet allocation table

Department	Network address	Subnet mask	Broadcast address	Valid host range	Usable hosts
School of Computing	192.168.100.0	/25 (255.255.255.128)	192.168.100.127	192.168.100.1 – 192.168.100.126	126
School of Electronics	192.168.100.128	/26 (255.255.255.192)	192.168.100.191	192.168.100.129 – 192.168.100.190	62
School of Mechanical Engineering	192.168.100.192	/27 (255.255.255.224)	192.168.100.223	192.168.100.193 – 192.168.100.222	30

Calculations

The number of usable hosts is calculated using:

$$\text{Usable hosts} = 2^{(32-\text{prefix length})} - 2$$

- **Computing (**/26**):** $2^7 - 2 = 126$usable hosts.
- **Electronics (**/26**):** $2^6 - 2 = 62$usable hosts.
- **Mechanical Engineering (**/27**):** $2^5 - 2 = 30$usable hosts.

Two addresses are subtracted for the network address and broadcast address.

Remaining unused IP address range

The allocated subnets occupy the following ranges:

- Computing: 192.168.100.0 – 192.168.100.127

- Electronics: 192.168.100.128 – 192.168.100.191
- Mechanical Engineering: 192.168.100.192 – 192.168.100.223

Therefore, the remaining unused range is:

192.168.100.224 – 192.168.100.255

This contains 32 IP addresses, from which 30 usable host addresses could be provided if allocated as a /27 subnet.

2. Software Company Network — 10.10.0.0/16

The given network block is 10.10.0.0/16. The subnets are allocated sequentially, starting from the beginning of the address block.

Subnet allocation table

Department	**Network address**	**Subnet mask**	**Broadcast address**	**Valid host range**	**Usable hosts**
Software Development	10.10.0.0	/24 (255.255.255.0)	10.10.0.255	10.10.0.1 – 10.10.0.254	254
Quality Assurance	10.10.1.0	/25 (255.255.255.128)	10.10.1.127	10.10.1.1 – 10.10.1.126	126
Human Resources	10.10.1.128	/26 (255.255.255.192)	10.10.1.191	10.10.1.129 – 10.10.1.190	62
Executive Management	10.10.1.192	/27 (255.255.255.224)	10.10.1.223	10.10.1.193 – 10.10.1.222	30

Calculations

Using the usable-host formula:

$$\text{Usable hosts} = 2^{(32-n)} - 2$$

- **Software Development (**/24**):** $2^8 - 2 = 254$
- **Quality Assurance (**/25**):** $2^7 - 2 = 126$
- **Human Resources (**/26**):** $2^6 - 2 = 62$
- **Executive Management (**/27**):** $2^5 - 2 = 30$

Remaining unused IP address range

The last allocated subnet ends at 10.10.1.223.

Therefore, the unused range is:

10.10.1.224 – 10.10.255.255

This range contains 65,056 IP addresses.

The original /16 network contains $2^{16} = 65{,}536$total addresses. The allocated subnets use:

$$256 + 128 + 64 + 32 = 480$$

Thus, the remaining number of addresses is:

$$65{,}536 - 480 = \boxed{65{,}056}$$

The unused range begins with 32 addresses in the remainder of 10.10.1.0/24, followed by all addresses in 10.10.2.0 through 10.10.255.255.

3. Smart Hospital IPv6 Network

Given IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

1. Compressed IPv6 notation

IPv6 addresses contain eight groups of hexadecimal digits. Leading zeros in a group can be omitted, and one consecutive sequence of zero groups can be replaced with ::.

The compressed notation is:

2001:db8:abcd::/64

2. Expanded hexadecimal representation

The full expanded address is:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

It contains eight groups, each representing 16 bits.

3. Network prefix

The prefix length is /64, meaning the first 64 bits identify the network.

- Network prefix: 2001:0db8:abcd:0000/64
- Network portion: first four hexadecimal groups
- Interface identifier: last four hexadecimal groups (64 bits)

The subnet can be represented as:

2001:db8:abcd:0000::/64

4–5. Total number of possible host addresses

IPv6 does not reserve network and broadcast addresses in the same way IPv4 does. A /64 subnet has 64 bits available for the interface identifier.

$$\text{Total addresses} = 2^{128-64} = 2^{64}$$

Therefore:

$$\boxed{18{,}446{,}744{,}073{,}709{,}551{,}616}$$

A /64 network supports 2^{64}possible interface identifiers, or approximately 18.45 quintillion addresses. Unlike IPv4, there is no need to subtract two addresses for network and broadcast addresses.

4. ISP Core Router — Routing Table

Given routing table entries:

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

Destination IP address: 192.168.2.45

1. Destination subnet

The destination address belongs to the subnet:

192.168.2.0/24

A /24 subnet covers addresses from 192.168.2.0 through 192.168.2.255.

2. Matching routing table entry

The matching entry is:

192.168.2.0/24

The destination IP address 192.168.2.45 falls within this range.

3. Next-hop network/interface

The router forwards the packet through the interface associated with the 192.168.2.0/24 route.

The exact outgoing interface or next-hop IP address cannot be determined from the supplied routing table alone. That information would be available in a complete routing table containing next-hop and interface details.

If the destination subnet is directly connected, the router forwards the packet through that LAN interface, using ARP to resolve the destination's MAC address when needed.

4. Default route

A default route is used when no more specific route matches the destination.

Its notation is:

0.0.0.0/0

Example configuration:

text

0.0.0.0/0 → ISP upstream next hop

The actual next-hop address must be supplied by the ISP or network configuration. The default route matches any IPv4 destination, but a more specific matching route takes precedence.

5. Administration and Finance LANs

Given subnet allocations:

- Administration LAN: 192.168.10.0/26
- Finance LAN: 192.168.10.64/26

Router gateway addresses:

- Administration: 192.168.10.1
- Finance: 192.168.10.65

A /26 subnet has 64 total addresses and 62 usable host addresses.

1–2. Network addresses, broadcast addresses, and host ranges

Detail	**Administration LAN**	**Finance LAN**
Network address	192.168.10.0	192.168.10.64
Subnet mask	255.255.255.192	255.255.255.192
Broadcast address	192.168.10.63	192.168.10.127
Valid host range	192.168.10.1 – 192.168.10.62	192.168.10.65 – 192.168.10.126
Usable hosts	62	62

3. Suggested IP addresses for three computers in each LAN

The following addresses are valid host addresses and do not conflict with the configured gateways.

Computer	Administration LAN	Finance LAN
Computer 1	192.168.10.2	192.168.10.66
Computer 2	192.168.10.3	192.168.10.67
Computer 3	192.168.10.4	192.168.10.68

Subnet mask for all six computers: 255.255.255.192.

4. Default gateway for each LAN

LAN	Default gateway
Administration	192.168.10.1
Finance	192.168.10.65

The default gateway is the router interface in the same subnet as the host. Computers use it when communicating with devices outside their local subnet.

For example, an Administration computer at 192.168.10.2 uses 192.168.10.1 to communicate with a Finance computer at 192.168.10.66. The router then forwards the traffic to the Finance LAN.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand